COM5231 · Introduction to Wireless and Ad Hoc Networking

Part 7 — Practice Questions

Generated practice questions with model answers

SUPPLEMENTARY content not found in the uploaded lecture notes

NOTES ERROR a mistake in the lecture notes, with the verified correction

EXAM how the topic is or could be examined

★ high-priority, frequently examined content

Exam preparation pack, built from the uploaded lecture materials and past papers

Part 7 — Additional Practice Questions with Model Answers

> **These are GENERATED practice questions, not taken from past papers.** They are built from your lecture notes and the patterns in Part 4, with high-frequency topics weighted most. Every numerical answer has been computed and checked. Suggested marks are shown in brackets.
>
> **How to use**: cover the answer, write your own answer in exam conditions, then mark it against the model.

SECTION A — Short definition questions (2 marks each)

A1. Define multiple access. *(2)* The technique that lets **many mobile users share a finite amount of radio spectrum simultaneously and efficiently**, so the system handles **as many calls as possible in a given bandwidth** (call-handling capacity) [MA p2].

A2. Define the path loss exponent. *(2)* A parameter **n** describing **how fast a radio signal's power decreases with distance** in a given environment ($Pr \propto 1/d^n$): 2 in free space, 2.7–3.5 in an urban cellular area, 4–6 obstructed in a building [WC-S 33–34].

A3. Define coherence time and state when a channel is fast fading. *(2)* **Coherence time Tc** is the time over which the channel stays relatively constant. The channel is **fast fading** when Tc is **not much greater** than the bit time Tb (Tc < Tb, Tc ≈ Tb or only slightly greater) [WC-B 5.3].

A4. What is intersymbol interference? *(2)* **Delayed copies of a pulse (from multipath) arriving at the same time as the primary pulse of a subsequent bit**, acting as noise to it and making detection difficult [WC-S 40].

A5. Define the NAV. *(2)* The **network allocation vector**: a timer set by stations that hear an RTS/CTS (which carry the duration of the exchange), showing **how long they must wait before checking the channel for idleness**. It is virtual carrier sensing [WL 37].

A6. What is a piconet? *(2)* The basic Bluetooth network unit: **one master and one to seven active slaves**, sharing the master's **frequency-hopping sequence and timing**. Slaves communicate only with the master, when permitted [BT-B].

A7. Define chip rate. *(2)* The **rate of a spreading code** in spread spectrum/CDMA. It is **much higher than the data bit rate** (e.g. 11 chips per bit with the Barker code), and is called the chip rate rather than the bit rate [MA p8; SS].

A8. What is the purpose of a guard band in FDMA? *(2)* An **unused frequency slot between adjacent channels** that **minimises crosstalk/adjacent-channel interference**, at the cost of wasted bandwidth [MA p4].

A9. Define a burst error and its length. *(2)* A burst error is **two or more bits changed** in a data unit. Its **length** is measured **from the first to the last corrupted bit, inclusive** (SUPPLEMENTARY).

A10. What is orthogonality in OFDM? *(2)* Two signals are orthogonal if **the average over a bit time of their product is zero**. For OFDM subcarriers spaced at multiples of 1/T, **each subcarrier's spectral peak falls where every other subcarrier's power is zero**, so they overlap without interfering [OF].

A11. What is a resource block in LTE? *(2)* The smallest group of subcarriers (and time slots) that OFDMA assigns to a user: **12 subcarriers** of 15 kHz [AT; OF].

A12. Name the three power classes of Bluetooth. *(2)* **Class 1**: 100 mW (+20 dBm), power control mandatory, greatest range. **Class 2**: 2.5 mW (+4 dBm). **Class 3**: 1 mW [BT-B].

SECTION B — Conceptual explanation questions

B1. Explain three reasons why CSMA/CD cannot be used in wireless LANs. (6)

1. **Simultaneous send and receive**: collision detection requires a station to transmit and listen at the same time. Received signals are much weaker than transmitted ones, so this needs costly full-duplex hardware and more bandwidth, and battery-powered stations lack the power [WL 16, 29; FZ].
2. **Hidden station problem**: a collision at the receiver may be **inaudible to the sender**, because the other transmitter is out of its range [FZ].
3. **Distance and fading**: stations can be far apart, so **signal fading** may stop a station at one end from hearing a collision at the other [WL 29]. Therefore 802.11 uses **CSMA/CA** with ACKs [FZ].

B2. Explain why forward error correction is preferred to backward error correction in wireless systems. (4)

- In backward error correction (ARQ) the receiver only detects errors and asks for a **retransmission**. In **satellite** links the **delay** makes this undesirable; in **mobile** links **error rates are so high that the retransmitted block is likely to be corrupted too** [WC-S 45].
- FEC adds redundant bits so the receiver **corrects errors itself**, using only the received data. This costs bandwidth (a total/data ratio of 2–3) but avoids retransmission delay [WC-B 5.4].

B3. Explain the role of the master's Bluetooth address in a piconet. (4) The master's address (and clock) determines the **frequency-hopping sequence (channel) and the phase (timing)** for all piconet members. Slaves tune to it [BT-B]. Because different masters have different addresses, **different piconets use different hop sequences**, which gives **FH-CDMA** separation in a scatternet. This is also why one device **cannot be master of two piconets**.

B4. Describe the three types of diversity and the two ways of using the received signals. (6) Space diversity: multiple antennas spaced far enough apart (or collocated directional antennas) experience independent fading. **Frequency diversity**: spread the signal over a wider band or multiple carriers (**OFDM, spread spectrum**). **Time diversity**: spread data over time so a noise burst hits fewer bits (**interleaving, Rake receiver**). Use: **selection diversity** (pick the best signal) or **combining diversity** (co-phase and add the signals) [WC-S 48; WC-B 5.4].

B5. Explain the exposed station problem and why RTS/CTS cannot solve it. (4) A transmits to B, and C wants to send to D. C's transmission would **not** interfere at B, but C is **within A's range**, hears A, and **refrains**: it is too conservative and **wastes capacity**. RTS/CTS doesn't help: C hears A's RTS and still defers, because **it cannot know that its transmission would not affect the zone between A and B** [FZ].

B6. Why does OFDM reduce the need for equalizers? (4) OFDM divides the data into **N parallel substreams**, so each subcarrier's **symbol time is N·Ts** instead of Ts. N is chosen so that N·Ts is **much greater than the channel's RMS delay spread**, so multipath delays occupy only a small fraction of each symbol and **ISI is drastically reduced**. Equalizers, which are complex and expensive and grow with ISI severity, **may not be needed** [OF §8.1].

B7. What is AODV and where is it used in your notes? (4) Ad hoc On-demand Distance Vector, a **reactive** MANET routing protocol. A source **sends route requests**, which nodes either **answer or forward** until someone (the destination or a node with a route) responds. Nodes **store and use** the routes they discover [BT-S 87]. Your notes mention it as the routing protocol **ZigBee** uses to build multi-hop ad hoc networks.

SECTION C — Comparison questions

C1. Compare FDMA and TDMA under: principle, synchronisation, efficiency, battery life, equalization. (10) | | FDMA | TDMA | |---|---|---| | Principle | Spectrum divided into **frequency bands with guard bands**; one user per channel **100% of the time** | Same frequency shared by dividing **time into slots**; users transmit in **bursts** | | Synchronisation | **Not severe** (continuous transmission) | **Required**: lost sync means slots collide | | Spectrum efficiency | **Inefficient**: idle channels can't be shared; guard bands waste bandwidth | **More efficient**: more users in the same spectrum | | Battery / handoff | Continuous transmission | **Longer battery life**; **simpler handoff** (listen in idle slots); but **high peak power** on the uplink | | Equalization | **Not required** | **Required** (high transmission rates) | | Duplexer | Needed (forward and reverse frequencies) | **Not required** (different slots for Tx/Rx) | | Examples | 1G AMPS, FM radio, walkie-talkies | GSM, IS-136, DECT | [MA p3–7]

C2. Compare ZigBee coordinator, router and end device. (6) | Device | Role | Notes | |---|---|---| | **Coordinator** | Most capable; **only one per network**; **starts the network**, is the root of a tree, **maintains network information such as security keys** | Usually mains powered | | **Router** | **Passes data on** to other devices (multi-hop) | Must stay awake in non-beacon networks | | **End device** | Just enough function to talk to a router/coordinator; **cannot relay** | **Sleeps most of the time**, least memory, **cheapest** | [BT-S 85–86]

C3. Compare SCO and ACL links. (6) | | SCO | ACL | |---|---|---| | Topology | Point-to-point (master ↔ one slave) | **Point-to-multipoint** (master ↔ all slaves) | | Slots | **Reserved at regular intervals** (2 consecutive, symmetric) | Any **unreserved** slot, per-slot polling | | Bandwidth | **Guaranteed** fixed rate | **No reservation** (packet-switched) | | Retransmission | **Never** | **Yes** (ARQ) | | Typical traffic | **Time-bounded voice** | **Data** | | Max number | Master: 3 SCO links | **One** ACL link | [BT-B]

C4. Compare ASK, BFSK and BPSK in terms of noise immunity and bandwidth. (6) **ASK**: data in amplitude; **most susceptible to noise** and sudden gain changes; bandwidth ∝ baud rate. **BFSK**: data in frequency; **less error-prone than ASK**; **needs more bandwidth** (two carriers separated by 2Δf). **BPSK**: data in phase (0°/180°); **less susceptible to noise than ASK** (noise changes amplitude more easily than phase); **same bandwidth as ASK, less than BFSK**; needs more sophisticated hardware [SE; WC-S 59–64].

SECTION D — Calculation and problem-solving questions

D1. Hamming code. Encode the data byte 10110011 using an even-parity Hamming code (positions numbered from 1 at the left, check bits at 1, 2, 4, 8). (a) How many check bits? (b) Calculate each check bit. (c) Write the codeword. (8) **(a)** $2^r \geq m + r + 1$ with m = 8: r = 3 gives 8 < 12 ✘; r = 4 gives 16 ≥ 13 ✔ → **4 check bits** (12-bit codeword). **(b)** | Pos | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | |---|---|---|

- C4 (4, 5, 6, 7, 12) = 1, 0, 1, 1, 1 → four 1s → OK → **0**
- C8 (8, 9, 10, 11, 12) = 0, 1, 0, 1, 1 → three 1s → fails → **1** Syndrome C8C4C2C1 = **1001 = 9** → **bit 9 is wrong** → flip it → **101101100011** ✔.

D3. CRC. Data 1101011011, generator $x^4 + x + 1$. Find the transmitted frame. (8)

- G = $x^4 + 0x^3 + 0x^2 + x + 1$ = **10011** (r = 4); append 0000 → **11010110110000**.
- Division (5-bit windows):
  1. 11010 ⊕ 10011 = 01001
  2. 10011 ⊕ 10011 = 00000
  3. 10110 ⊕ 10011 = 00101
  4. 10100 ⊕ 10011 = 00111
- **Remainder = 1110** → **transmitted frame = 1101011011 1110**.
- Check: dividing 11010110111110 by 10011 gives remainder 0000 ✔.

D4. CRC. Data 100100, generator 1101. Find the CRC and the transmitted string. (6) Append 000 → 100100000.

1. 1001 ⊕ 1101 = 0100
2. 1000 ⊕ 1101 = 0101
3. 1010 ⊕ 1101 = 0111
4. 1110 ⊕ 1101 = 0011
5. 1100 ⊕ 1101 = 0001 **Remainder = 001** → transmit **100100001**.

D5. Internet checksum (4-bit words). Message 0110 1011 1001 0101. Find the checksum and show the receiver's check. (4) | Step | Addition | Raw | After end-around carry | |---|---|---|---| | 1 | 0110 + 1011 | 1 0001 | 0001 + 1 = **0010** | | 2 | 0010 + 1001 | 1011 | **1011** | | 3 | 1011 + 0101 | 1 0000 | 0000 + 1 = **0001** | **Sum = 0001 → checksum = 1110.** Receiver: 0001 + 1110 = **1111** → no error. (Decimal check: 6 + 11 + 9 + 5 = 31 = 2 × 15 + 1 → 0001 ✔.)

D6. Free-space received power. A 2.4 GHz access point transmits 100 mW. Find the received power at 50 m in watts and dBm. (6)

- $\lambda = c/f = 3 \times 10^8 / 2.4 \times 10^9$ = **0.125 m**
- $4\pi d/\lambda = 4\pi \times 50 / 0.125$ = **5026.5** → squared = **2.527×10^7**
- $P_r = P_t / (4\pi d/\lambda)^2 = 0.1 / 2.527 \times 10^7$ = **3.96×10^{-9} W (3.96 nW)**
- In dBm: $10 \log(3.96 \times 10^{-9} / 10^{-3})$ = **−54.0 dBm** (Equivalently, L_dB = 20 log f + 20 log d − 147.56 = 187.60 + 33.98 − 147.56 = 74.0 dB, and 20 dBm − 74 dB = −54 dBm ✔.)

D7. Shannon capacity. A 20 MHz channel has SNR = 30 dB. Find the Shannon capacity. How many signal levels would Nyquist require for this rate? (6)

- SNR = 10^(30/10) = **1000**
- $C = B \log_2(1 + SNR) = 20 \times 10^6 \times \log_2(1001) = 20 \times 10^6 \times 9.967$ = **199.3 Mbps** (≈ 200 Mbps)
- Nyquist: $199.3 \times 10^6 = 2 \times 20 \times 10^6 \times \log_2 M$ → $\log_2 M = 4.98$ → **M ≈ 32 levels**

D8. FDMA channels. A system has Bt = 25 MHz, guard bands Bg = 100 kHz at each edge, Bc = 200 kHz. How many channels? If each channel carries 8 TDMA slots, how many users? (4)

- N = (Bt − 2Bg)/Bc = (25 000 − 200)/200 = 24 800/200 = **124 channels**
- Users = 124 × 8 = **992** [MA p4, p6]

D9. Scatternet probability. Three piconets share 80 frequencies (always transmitting, no CSMA, independent hops). What is the probability that a poll + reply transaction in piconet 1 suffers a

collision? (6)

- In one slot, piconet 1 avoids collision only if **both** other piconets choose a different frequency: $(79/80)^2$ per slot.
- The transaction needs 2 slots: P(success) = $((79/80)^2)^2 = (79/80)^4$ = **0.9509**.
- **P(collision) = 1 − $(79/80)^4$ ≈ 0.049 ≈ 4.9%**. Adding piconets increases collisions, as the notes say [BT-B].

D10. Dijkstra. Using Figure 1 (the exam network), compute the shortest paths from u to all nodes. (8)
Link costs: x–z 8, x–y 6, x–v 3, x–w 6, y–z 12, y–v 8, y–t 7, v–t 4, v–u 3, v–w 4, t–u 2, w–u 3.

Step	N'	D(t),p	D(v),p	D(w),p	D(x),p	D(y),p	D(z),p
0	u	**2,u**	3,u	3,u	∞	∞	∞
1	ut	—	**3,u**	3,u	∞	9,t	∞
2	utv	—	—	**3,u**	6,v	9,t	∞
3	utvw	—	—	—	**6,v**	9,t	∞
4	utvwx	—	—	—	—	**9,t**	14,x
5	utvwxy	—	—	—	—	—	**14,x**
6	utvwxyz	—	—	—	—	—	—

Updates: via t: y = 2 + 7 = 9 (v via t = 6 > 3). Via v: x = 3 + 3 = 6 (y via v = 11 > 9; w via v = 7 > 3). Via w: x via w = 9 > 6. Via x: z = 6 + 8 = 14 (y via x = 12 > 9). **Results**: t 2 (u-t); v 3 (u-v); w 3 (u-w); x 6 (u-v-x); y 9 (u-t-y); z 14 (u-v-x-z).

D11. Bellman-Ford. Node x has neighbours y (cost 2) and z (cost 7). y advertises D_y(z) = 1 and D_y(w) = 5; z advertises D_z(w) = 3. Compute D_x(z) and D_x(w). (4)

- D_x(z) = min{ c(x,y) + D_y(z), c(x,z) + D_z(z) } = min{2 + 1, 7 + 0} = **3 (via y)**
- D_x(w) = min{ c(x,y) + D_y(w), c(x,z) + D_z(w) } = min{2 + 5, 7 + 3} = **7 (via y)**

D12. Path loss. A 900 MHz cellular link has a receiver 2 km from the base station. Find the free-space path loss as a ratio and in dB. (4)

- $\lambda = 3 \times 10^8 / 9 \times 10^8$ = **0.333 m**
- $L_p = (4\pi d/\lambda)^2 = (4\pi \times 2000 / 0.333)^2 = (75\,398)^2$ = **5.68×10^9**
- $L_{dB} = 10 \log(5.68 \times 10^9)$ = **97.5 dB** (check: $20 \log(9 \times 10^8) + 20 \log(2000) - 147.56 = 179.08 + 66.02 - 147.56 = 97.5$ ✔)

SECTION E — Application and scenario questions

E1. A university library's WiFi slows down every lunchtime, and users report many retransmissions. The APs use 2.4 GHz channel 6. A staff kitchen with microwave ovens is next door. Diagnose and recommend. (10)

- **Diagnosis**: microwave ovens radiate around **2.45 GHz** in the same **ISM band** as 2.4 GHz WiFi [WCP §10]. At lunchtime the ovens cause **inward/co-channel interference**, which **raises the bit error rate**. Corrupted frames are not ACKed, so stations retransmit and back off (CSMA/CA), and throughput falls [WL 27, 33]. ISM users have **no regulatory protection** [WC-S 10].
- **Recommendations**: (1) move the APs to the **5 GHz band** (less crowded, no ovens) [WL 23]; (2) change the 2.4 GHz channel to one away from the oven's emissions (e.g. channel 1); (3) **relocate APs** away from the kitchen or add **shielding**; use **directional antennas** pointing away; (4) **add APs /**

reduce cell size so clients have a stronger signal (higher SINR); (5) allow **rate adaptation** and **fragmentation** (smaller frames are less likely to be hit) [FZ]; (6) **verify** with a spectrum analyser before and after.

E2. Five laptops in a meeting room with no AP want to share files. Which 802.11 architecture applies? What limitation do they face? (4) An **ad hoc BSS (independent BSS)**: stations form a network **without an AP**, locating each other and agreeing to be part of a BSS [FZ]. **Limitations**: it is a **stand-alone network that cannot send data to other BSSs** or the Internet [WL 10; FZ]. Stations communicate only if **within each other's range** (or if others forward) [WL 9]. PCF is not available (infrastructure only) [WL 40].

E3. A pedestrian (Tc ≈ 70 ms) and a car on the highway (Tc ≈ 5 ms) both use a 1 kbps telemetry link. Classify the fading for each. (4) Bit time Tb = 1/1000 = **1 ms**.

- **Pedestrian**: Tc = 70 ms ≫ 1 ms → **slow fading** (the channel hardly changes during a bit).
- **Car**: Tc = 5 ms, which is only 5× Tb, **not much greater** → by the notes' definition it is treated as **fast fading** [WC-B 5.3].

E4. A mobile user moves to the cell edge in an LTE network. Using your OFDMA notes, explain how the scheduler can still serve this user fairly. (6) Pure efficiency scheduling would favour users with the **highest SINR**, starving cell-edge users [OF]. **Proportional fairness** compares each user's **achievable rate now (r_i) to its own average rate ($\bar{r}_i$)**: the metric is $r_i/\bar{r}_i$. When the edge user's channel momentarily improves **relative to its own average**, its metric rises and it is scheduled, even if its absolute rate is lower than central users'. Subchannel choice (adjacent blocks matched to good frequencies), **AMC** and higher power on its subchannels also help. In the uplink, **subchannelization concentrates the device's power** on its allocated subchannels [OF §8.2].

SECTION F — Long-form examination questions

F1. "Wireless channels are hostile." Discuss the main impairments of the wireless channel and the mechanisms modern systems use to overcome them. (20) *Model answer outline (≈ 1 mark per developed point):*

1. **Introduction (2)**: the wireless channel is shared, time-varying and unpredictable. The received signal differs from the transmitted one because of impairments, causing random degradation (analog) or bit errors (digital) [WC-B 5.2].
2. **Impairments (9)**:
   - **Free-space loss/path loss**: $(4\pi d/\lambda)^2$; path loss exponent n [WC-S 30–34].
   - **Noise**: thermal (irreducible), intermodulation, crosstalk, impulse (the main digital error source) [WC-S 36–38].
   - **Atmospheric absorption and refraction** [WC-S 39].
   - **Multipath** via reflection, diffraction and scattering → **destructive addition (20–30 dB)** and **ISI** [WC-B 5.3].
   - **Fading**: large-scale (shadowing) and small-scale; **fast/slow** (coherence time) and **flat/frequency-selective** (coherence bandwidth); Rayleigh/Rician [WC-B 5.3].
   - **Interference**: co-channel and adjacent channel; the unprotected ISM band [WCP §10].
3. **Mechanisms (8)**:
   - **FEC** (redundancy; ARQ is impractical) [WC-S 45].
   - **Adaptive equalization** (combats ISI) [WC-S 47].
   - **Diversity** (space, frequency, time; selection or combining) [WC-S 48].

- **MIMO**: diversity, beamforming, MU-MIMO, multilayer [WC-B 5.4].
- **AMC** (adapt hundreds of times per second) [WC-S 51].
- **OFDM** (long symbols, selective fading on few subcarriers) [OF].
- **Spread spectrum** (100× bandwidth, anti-interference) [SS].
- **Bandwidth expansion** (carrier aggregation, reuse, mmWave) [WC-S 52].

4. **Conclusion (1)**: real systems **combine** all of these (e.g. LTE and 802.11ac use OFDM + MIMO + AMC + FEC) [WC-B 5.4].

F2. Discuss medium access control in IEEE 802.11. Include the reason CSMA/CD is not used, DCF with RTS/CTS and NAV, the role of IFS, PCF, and the hidden and exposed station problems. (20) *Outline*:

- MAC functions: reliable delivery, access control, security [WL 27] (2)
- Why not CSMA/CD: three reasons [WL 29] (3)
- DCF/CSMA/CA steps: sense, DIFS, backoff (freeze), RTS/CTS/DATA/ACK with SIFS, no ACK → backoff and retry [WL 32–35] (5)
- NAV and collisions during handshaking [WL 37–38] (2)
- IFS: SIFS < PIFS < DIFS and the priority each gives [WL 34] (2)
- PCF: AP polling, PIFS priority, infrastructure only, superframe, single point of failure [WL 40–42] (3)
- Hidden station (CTS solves it) and exposed station (not solved) [FZ] (3)

F3. Bluetooth is described as robust, low-power and low-cost. Explain how its architecture and protocols achieve these goals. (15) *Outline*:

- **Robustness**: FHSS at 1600 hops/s over 79 channels (resists interference and multipath); adaptive FH; 1/3 and 2/3 FEC; CRC + ARQ; HEC; SEQN duplicate filtering [BT-B; BT-S 45–50] (5)
- **Low power**: short range (10 m), power classes and **power control via LMP**, TDD on one chip, BLE (10 mW, long battery life) [BT-B; BT-S 53–55] (5)
- **Low cost**: one-chip transceiver (TDD prevents crosstalk), unlicensed ISM band, reuse of adopted protocols (PPP, TCP/IP, OBEX), profiles so devices implement only what they need, single-mode BLE chips [BT-B] (5)

F4. Discuss routing in mobile ad hoc networks: why it is hard, the three categories of protocols with examples, and how distance-vector problems such as count-to-infinity are avoided in DSDV and AODV. (20) *(SUPPLEMENTARY) Outline*: challenges (6: mobility, no infrastructure, bandwidth, energy, unreliable links, security); proactive (4: DSDV, OLSR); reactive (4: AODV, DSR; RREQ/RREP/RERR); hybrid (3: ZRP); loop avoidance (3): **destination sequence numbers** in DSDV and AODV mark route freshness, so a node never accepts an older (stale) route. This prevents the stale-route loops behind count-to-infinity.